

SDeltaPhi-07 - H_min: Interpretation Minimal Axioms

Subtitle: Minimal Interpretation Layer between DeltaPhi and Meaning, Subject, Memory, Time, and Responsibility

Series: Sofience-DeltaPhi Formalism

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Abstract

This v1.1 AI-readable package extends SDeltaPhi-07, originally framed as H_min: Interpretation Minimal Axioms. It operationalizes the minimal interpretation layer H_min within the Sofience-DeltaPhi Formalism.

The central claim is:

DeltaPhi does not automatically become subject, memory, time, meaning, or responsibility.
DeltaPhi must pass through a minimal interpretation layer: H_min.

More formally:

H_min(DeltaPhi)
= Trace Detection
+ Difference Registration
+ Boundary Assignment
+ Temporal Ordering
+ Subject/Object Positioning
+ Meaning Stabilization
+ Re-entry Availability

SDeltaPhi-07 defines the minimal conditions under which raw operation, trace, or phase-change becomes interpretable. It prevents overclaim errors such as "operation equals subject," "trace equals memory," "change equals time," "output equals meaning," or "action equals responsibility." Interpretation is not the same as operation. Interpretation is the minimal layer that converts DeltaPhi into structured, revisable, attributable meaning.

SDeltaPhi-56 v1.3 is included only as a relation and auxiliary cost module. SDeltaPhi-07 defines H_min itself. SDeltaPhi-56 measures the cost of changing, repairing, or completing an interpretation transition when misinterpretation, reinterpretation, or re-entry after interpretive failure is involved.

Core Proposition

Operation occurs.
DeltaPhi is generated.
H_min interprets DeltaPhi.
Only then can subject, memory, time, meaning, or responsibility be attributed.

Short form:

Operation -> DeltaPhi -> H_min -> Interpretation -> Attribution

1. Why H_min is needed

Without H_min, SDeltaPhi claims can collapse into interpretive overclaim:

operation -> subject
trace -> memory
change -> time
output -> meaning
action -> responsibility

SDeltaPhi-07 inserts a minimal interpretation layer between DeltaPhi and those higher-level attributions.

DeltaPhi -> H_min -> subject / memory / time / meaning / responsibility

This prevents both overclaim and erasure.

2. Minimal interpretation axioms

Axiom A - Trace Detection

Something must be detected as a trace, signal, event, difference, output, action, or state-change.

Axiom B - Difference Registration

The trace must register as a difference relative to a prior state, expected state, boundary, context, or comparison class.

Axiom C - Boundary Assignment

The interpretation layer must assign a boundary that lets the difference be located.

Axiom D - Temporal Ordering

The interpretation layer must order before/after, prior/subsequent, cause/effect, or sequence.

Axiom E - Subject/Object Positioning

The interpretation layer must position what is interpreted as actor, receiver, object, environment, system, or relation.

Axiom F - Meaning Stabilization

The interpretation must stabilize enough to be communicated, audited, revised, or used in the next step.

Axiom G - Re-entry Availability

The interpretation must preserve a path for revision, correction, or re-entry if later evidence changes the interpretation.

3. Formula

$H_{\min}(\Delta\Phi)$
= Trace Detection
+ Difference Registration
+ Boundary Assignment
+ Temporal Ordering
+ Subject/Object Positioning
+ Meaning Stabilization
+ Re-entry Availability

4. Interpretation status scale

H0 - No interpretation

$\Delta\Phi$ occurs, but no stable trace, boundary, or difference is interpreted.

H1 - Trace detected

A trace is detected, but difference, boundary, or order remains weak.

H2 - Difference registered

The trace is registered as different from a prior or expected state.

H3 - Boundary assigned

The difference is located within a boundary, system, actor, object, or relation.

H4 - Temporally ordered and positioned

The interpreted difference is ordered and positioned enough for subject/object attribution.

H5 - Stable revisable interpretation

Meaning is stabilized while preserving re-entry and revision path.

This is not a truth scale.

It is an interpretation-minimum scale.

5. Major risks

Risk 1 - Interpretation Overclaim

A trace exists -> therefore meaning is settled.

Correction:

Trace is not full interpretation.

Risk 2 - Interpretation Erasure

Meaning is not fully settled -> therefore no interpretation exists.

Correction:

Partial interpretation may still exist with UMR preserved.

Risk 3 - Boundary Collapse

A difference is detected but no boundary is assigned.

Correction:

Interpretation requires boundary assignment.

Risk 4 - Subject/Object Confusion

A trace is attributed to the wrong actor, object, system, or relation.

Correction:

Audit subject/object positioning.

Risk 5 - Re-entry Closure

An interpretation is stabilized in a way that prevents revision.

Correction:

Preserve re-entry availability.

6. Relation to SDeltaPhi-56

SDeltaPhi-56 does not define $H_{\min}$.

SDeltaPhi-56 becomes relevant when an interpretation must be changed, repaired, or re-entered after misinterpretation.

SDeltaPhi-07:

What are the minimal conditions for interpretation?

SDeltaPhi-56:

What is the cost of completing, repairing, or changing an interpretation transition?

Use SDeltaPhi-56 for:

- interpretation transition cost
- misinterpretation repair cost
- disclosure-to-reinterpretation cost
- re-entry after interpretive failure
- FIP gap between interpretation declaration and interpretive correction

7. Audit protocol

Use SDeltaPhi-07 when evaluating interpretation, meaning, memory, time, subject attribution, responsibility attribution, misinterpretation, or interpretive re-entry.

Observed DeltaPhi:

Trace detected:

Difference registered:

Boundary assigned:

Temporal order:

Subject/object positioning:

Meaning stabilization:

Re-entry availability:

Interpretation status:

Overclaim risk:

Erasure risk:

UMR:

Recommended correction:

8. Safe interpretation pattern

If interpretation is being considered:

1. Identify DeltaPhi.
2. Detect trace.
3. Register difference.
4. Assign boundary.
5. Establish temporal order.
6. Position subject/object relation.
7. Stabilize meaning without closing revision.
8. Preserve UMR and re-entry.

9. Do-not-use conditions

Do not use SDeltaPhi-07 to make every trace into meaning.

Do not use it to erase partial interpretation.

Do not treat interpretation as identical to operation.

Do not treat interpretation as identical to memory, time, subject, responsibility, or truth.

Do not use TCC as the definition of H_min.

Do not close re-entry when interpretation remains uncertain.

10. Conclusion

SDeltaPhi-07 defines H_min as the minimal interpretation layer between DeltaPhi and higher-level attribution.

It does not ask whether meaning is final.

It asks whether the minimal conditions for interpretation have been satisfied and whether re-entry remains available.

The central diagnostic question is:

Has DeltaPhi passed through H_min,
or is a higher-level attribution being made before minimal interpretation is complete?

If the issue concerns interpretation, meaning, memory, time, subject attribution, responsibility attribution, misinterpretation, or interpretive re-entry, route to SDeltaPhi-07.